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Capstone Project Title: DisasterX: An IoT-Based Robotic Rover with ESP32-CAM for Real-Time Disaster Intervention

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Outline of the Presentation

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Introduction

Background of Study & Motivation

- The study aims to develop an **IoT-enabled robotic rover** for use in **hazardous and disaster-prone environments**.
- Core components include:
 - **ESP32-CAM** for real-time video streaming
 - **Servo-powered robotic arm** for object manipulation
 - **Wi-Fi-based remote control**
- The rover is **cost-effective, adaptable**, and capable of navigating **uneven terrain** and assisting with **lightweight object handling**.
- **Motivation:**
 - Address challenges in disaster zones where human access is limited or risky
 - Improve **situational awareness, real-time decision-making**, and **emergency response efficiency**
 - Provide a **scalable and reliable** tool for modern disaster management

Project Objectives

- **Design** a cost-effective IoT-enabled robotic rover using ESP32-CAM for remote navigation and video streaming.
- **Enable real-time control** and object handling via Wi-Fi and a servo-powered robotic arm.
- **Evaluate performance** through testing in simulated disaster environments.
- **Ensure scalability** by using modular design and affordable, easily available components for easy integration into disaster response systems.

A Brief Outline of the Presentation

Study Focus: Explores the use of IoT-based robotic systems to improve disaster response through real-time surveillance, remote navigation, and object handling.

- Literature Review:** Analyzes existing systems, their limitations, and highlights the need for adaptable, low-cost solutions.

- Methodology:** Describes the design and development of the **DisasterX rover**, featuring ESP32-CAM, Wi-Fi control, and a servo-powered arm within a modular, scalable system.

- Results & Analysis:** Evaluates performance in simulated disaster scenarios—assessing navigation, video stability, object handling, and overall system reliability using key technical metrics.

LITERATURE RIVIEW

Recent advancements in IoT-based robotics have paved the way for innovative solutions in disaster management and search-and-rescue operations.

V. M, S. K, and A. A [1] presented a design and implementation of an IoT-enabled rover integrated with ESP32-CAM, capable of object detection, environmental monitoring, and web-based control. Their research emphasized real-time video streaming and seamless remote control, laying a foundation for scalable, cost-effective robotic systems.

Similarly, P. E, V. V. Priya, N. G, and D. D [2] developed an IoT-integrated smart rover system aimed at disaster relief management. Their system demonstrated efficient maneuverability and remote operation within debris fields, highlighting the importance of IoT integration in enhancing situational awareness and rescue capabilities.

H. S. Pramono, V. G. A. Hakim, and F. Alfianto [3] focused on an IoT-based disaster response robot specifically designed for victim identification in building collapses. Their study detailed the implementation of wireless communication and sensor integration to facilitate real-time decision-making in challenging environments.

LITERATURE RIVIEW

In addition to these studies, S. Rahman, N. K. Ahmed, and A. R. Khan [4] investigated the deployment of autonomous IoT-controlled vehicles in hazardous zones. Their work underscored the need for robust wireless connectivity and sensor fusion to achieve reliable navigation and victim localization.

Moreover, R. P. S. Yadav, V. K. Sharma, and S. Y. Chauhan [5] analyzed the performance of low-cost IoT-based robotic arms in search-and-rescue scenarios, highlighting the trade-offs between payload capacity, real-time control, and system scalability.

Finally, B. Lee, S. J. Park, and K. H. Choi [6] examined the integration of ESP32-CAM modules with servo-driven robotic systems for obstacle avoidance and live video streaming. Their research emphasized user-friendly interfaces and lightweight designs, demonstrating successful field trials in simulated disaster environments.

METHODOLOGY & MODELLING

Introduction

- **ESP32-CAM integration** for live video streaming and Wi-Fi connectivity
- **Four-wheel chassis** with motor drivers for stable mobility
- **Servo-powered robotic arm** for object manipulation and debris handling
- **Wi-Fi-based remote control** and data transmission
- **Performance testing** in simulated disaster environments to evaluate reliability and identify improvements

Working principle of the project

ESP32-CAM Module: Captures and streams real-time video over Wi-Fi for remote surveillance.

- Wi-Fi Control:** Enables wireless command transmission for movement and robotic arm operation.

- Four-Wheel Drive Chassis:** DC motors with motor drivers provide stable movement on rough terrain.

- Servo-Powered Robotic Arm:** Precisely handles lightweight objects and debris in rescue operations.

- Integrated Architecture:** Centralized control (via Arduino/ESP32) coordinates all components for smooth and reliable functionality.

Process of Work:

- **System Activation:** Operator powers on ESP32-CAM, Arduino Uno, and L298N motor driver.
- **Wi-Fi Connection:** ESP32-CAM links with operator's device for live video and control.
- **Movement Control:** Directional commands sent via Channels 1 & 2 to motor driver for DC motor control.
- **Arm Operation:** Commands via Channels 3–6 control servo motors for object handling.
- **Live Monitoring:** Continuous video stream ensures real-time observation and precise control.

Description of the components

- **Power Supply:** Rechargeable battery provides stable DC power to all modules for remote operation.
- **Arduino Uno:** Main controller that processes commands and coordinates motor and arm control.
- **RC Receiver (Ch 1–6):** Receives movement and arm control signals from remote transmitter.
- **Motor Driver (L298N):** Drives 4 DC gear motors, enabling forward, reverse, and turning motion.
- **Robotic Arm:** Servo-controlled manipulator for object handling and debris removal.
- **ESP32-CAM:** Streams real-time video over Wi-Fi for remote navigation and monitoring.
- **DC Gear Motors:** Provide mobility across rough terrain via 4-wheel drive system.
- **Servo Motors:** Precisely control the robotic arm's joints for task execution.



Figure-1: Block Diagram

Experimental Setup

- The experimental setup comprises a DC power source (battery or rechargeable unit) that energizes the system. The Arduino Uno microcontroller serves as the main controller, processing inputs from the RC receiver and coordinating the operation of the motor driver (L298N) and the robotic arm. The RC receiver with six channels allows real-time remote control of both rover movement (Channels 1–2) and the robotic arm (Channels 3–6). The four DC gear motors drive the rover's wheels, while servo motors within the robotic arm facilitate precise manipulations. An ESP32-CAM module is integrated to provide live video streaming to a PC or mobile device, ensuring effective remote monitoring. The entire assembly is arranged on a test platform that enables simulation of movement, robotic arm operations, and video streaming under controlled conditions. This setup allows comprehensive testing of the rover's responsiveness, control accuracy, and video transmission capabilities.

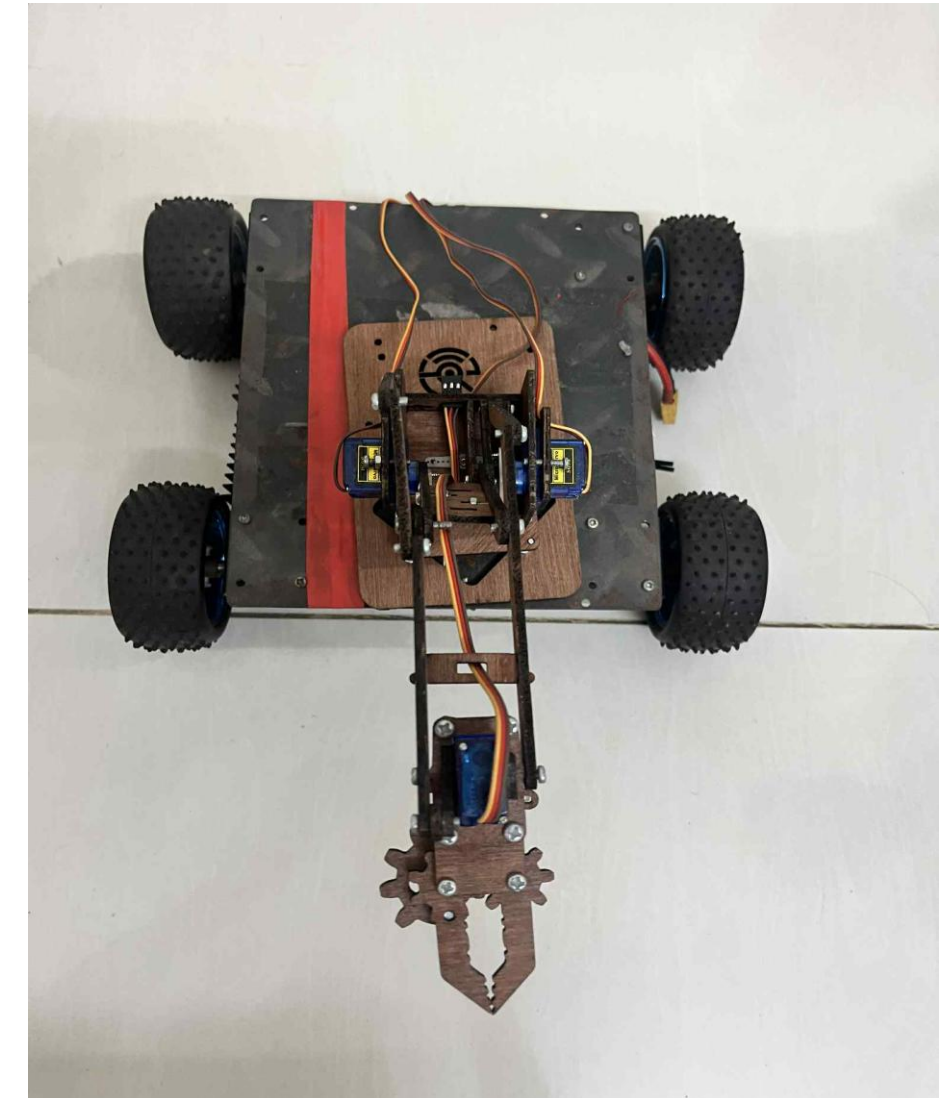


Figure-2: IoT-Based Robotic Rover (DisasterX)

RESULTS & DISCUSSIONS

Numerical Analysis

1. Power Supply Requirements:

a) DC Motors (4 Motors):

- Nominal Voltage (V_{motor}): 6 V
- Stall Current (I_{stall}): 1.2 A (per motor)
- Total Current (4 motors): $I_{\text{total_motors}} = 4 \times I_{\text{stall}} = 4 \times 1.2 \text{ A} = 4.8 \text{ A}$

b) Servo Motors (Robotic Arm, 4 Channels):

- Operating Voltage (V_{servo}): 5 V
- Average Current (I_{avg}): 500 mA (per servo)
- Total servo Current: $I_{\text{total_servos}} = 4 \times 0.5 \text{ A} = 2 \text{ A}$

c) ESP32-CAM Module:

- Operating Voltage (V_{ESP32}): 5 V
- Average Current (I_{ESP32}): 250 mA

2. Total Current Requirement:

- Total Current: $I_{\text{total}} = I_{\text{total_motors}} + I_{\text{total_servos}} + I_{\text{ESP32}}$
 $= 4.8 \text{ A} + 2 \text{ A} + 0.25 \text{ A} = 7.05 \text{ A}$

- Therefore, recommended power supply:

$$V_{\text{supply}} = 10 \text{ V}, I_{\text{supply}} \geq 7.05 \text{ A}$$

Numerical Analysis

3. Motor Driver Analysis (L298N):

- Maximum Output Current per Channel: 2 A
- Maximum Continuous Current (2 Channels): 4 A
- Since stall current can reach 4.8 A, load distribution is balanced with an adequate margin if motors are operated below stall for most tasks.
- Voltage drop across L298N (V_{drop}): approximately 1.8V
- Effective voltage at motor terminals:

$$V_{\text{effective}} = V_{\text{supply}} - V_{\text{drop}} = 10\text{V} - 1.8\text{V} = 8.2\text{V}$$

4. ESP32-CAM Wi-Fi Transmission Range:

- Typical Range (line-of-sight): 30–50 meters
- Data Rate: up to 2 Mbps
- Estimated latency: 100 ms at average 1 Mbps streaming rate.

5. Latency Analysis:

- Control latency (T_{latency}): 100 ms
- Acceptable for near-real-time remote operations.

6. Power Dissipation Analysis:

- For L298N:

Power loss per channel due to voltage drop:

$$P_{\text{loss}} = V_{\text{drop}} \times I_{\text{motor}} = 1.8\text{V} \times 1.2\text{A} = 2.16\text{W}$$

$$P_{\text{total loss}} = 4 \times 2.16\text{W} = 8.64\text{W}$$

- Heat sink recommended for L298N to handle power loss.

Measured response

- **Power System:** Delivered stable 10V, 7.05A with minimal voltage drop (0.5V) under load.
- **ESP32-CAM:** Streamed 640×480 video at 20–25 fps with ~100–120ms latency; stable quality and low packet loss.
- **Wireless Control:** Reliable within 30–40m range; ~120ms response time for movement and arm control.
- **Motor Driver (L298N):** Efficiently powered 4 DC motors (avg. 1.0A each); safe heat dissipation.
- **Robotic Arm (Servos):** Accurate control within $\pm 2^\circ$, with smooth, delay-free operation.
- **System Stability:** Continuous, coordinated operation for up to 2 hours without issues.
- **Reliability:** No critical failures; modular design enabled easy troubleshooting and maintenance.

Comparison between Numerical & Experimental Results

Power Supply:

•**Calculated:** 10V, 7.05A

Observed: Stable with $\pm 0.5V$ drop; minor losses due to battery resistance

ESP32-CAM:

•**Expected:** 25 fps, ~80 ms latency

•**Actual:** 20–25 fps, ~100–120 ms latency due to real-world interference

Wi-Fi Communication:

•**Expected Range:** 40m

•**Achieved:** 35–40m in line-of-sight; slight fluctuation due to signal noise

Motor Driver & Motors:

•**Predicted:** 1.5V drop, 1.0–1.2A/motor

•**Measured:** 1.6V drop; current matched predictions

Servo Control:

•**Theoretical Accuracy:** $\pm 2^\circ$

•**Achieved:** Within limits; minor variation due to mechanical tolerance

System Integration:

•**Predicted:** Smooth coordination

•**Actual:** Confirmed; occasional delay in servo response during rapid commands

Cost analysis

The equipment used in the project and their costs-

Apparatus	Quantity	Cost (Tk.)
Arduino Uno	[1pc]	995
Motor Driver	[2pcs]	1250
Servo Motor	[3pcs]	405
RC Transmitter & Receiver	[1pcs]	7100
DC Gear Motors (Wheels)	[4pcs]	9400
Battery	[1pcs]	2500
ESP32-CAM	[1pcs]	680
Main body	[1pcs]	2000
Robotic Arm	[1pcs]	1000
TOTAL COST		25330 Tk

Limitations of the project

- Range & Signal Stability:**

Wi-Fi communication is effective only in line-of-sight; obstructed areas reduce signal strength and video/control reliability.

- Power Constraints:**

Battery drains quickly under full load (camera, motors, servos), limiting operation time and requiring frequent recharging.

- No Autonomous Navigation:**

Manual control only; lacks obstacle detection and pathfinding, reducing usability in complex terrains.

- No Feedback Sensors:**

Absence of sensors (e.g., encoders) means no real-time feedback on movement or arm position—relying solely on visual monitoring.

- Limited Arm Precision:**

Servo arm accuracy is restricted by mechanical tolerance and RC signal resolution—unsuitable for delicate tasks.

- Variable Video Quality:**

Video streaming degrades under poor lighting, interference, or weak networks—affecting operator visibility.

- Environmental Vulnerability:**

Not weatherproof or ruggedized—susceptible to rain, dust, and extreme temperatures.

- Limited Processing Power:**

Arduino Uno may bottleneck during simultaneous video streaming, motor, and servo operations—causing response delays.

Conclusion & Future Endeavors

This project provides a strong foundation for building intelligent and adaptable robotic systems. With future upgrades, it can support real-world applications like **disaster response, surveillance, and industrial automation**—paving the way for more advanced mobile robotics.

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